Understanding time or fitting patterns? A self-consistency test for time-series foundation models

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Task question (Topic 4, "A foundation model that truly understands time"): What would it mean for a model to truly understand temporal dynamics rather than only fit temporal patterns? How would you build one, and how would you know it had succeeded?

1 Central claim

Understanding how a series evolves, rather than fitting its patterns, should at least mean that all of a forecaster's predictions come from a single probabilistic model of the series. I turn this capability into a test that uses only the forecaster's own predictions and needs no ground truth. By the law of total probability, the direct forecast h steps ahead must equal the *chained* forecast: predict one step, append each possible next value to the history, predict the remaining steps and average. Good accuracy at each horizon does not guarantee this, because a forecaster that is optimal at every horizon separately can still fail. Failing shows that the forecasts cannot all come from one model. Passing does not prove understanding of time: a forecaster that feeds its own one-step predictions back in passes automatically. I applied the test to six pretrained time-series foundation models. All of them fail it on every series tested, by a small but systematic margin.

2 The project

2.1 The capability and the test

A forecaster maps a history $c = (y_1, \ldots, y_t)$, the *context*, to a forecast distribution $q_h(\cdot \mid c)$ of y_{t+h} for each horizon h. Time-series foundation models (TSFMs) are pretrained on many series. Many of them, including the six I test, output a forecast for every horizon up to 64 in a single pass; I call these the *direct* forecasts.

Suppose a forecaster has a single probabilistic model of the process: all its forecasts are conditional distributions of one joint distribution over future paths, given the history.[1] Then the law of total probability, applied to the next value, links its horizons. For every context and every $h \geq 2$,

$$q_h(\cdot \mid c) \;=\; T_h(\cdot \mid c) \;:=\; \int q_{h-1}(\cdot \mid c \oplus y)\; q_1(\mathrm{d}y \mid c), \tag{1}$$

where $c \oplus y$ is the context with y appended and nothing removed. I call T_h the *chained forecast*: predict one step, append each possible next value y, predict the remaining $h - 1$ steps, and average over y using its one-step probability (Figure 1).

Example 1 (A first-order autoregressive process)**.** Let $y_{t+1} = \phi y_t + \varepsilon_{t+1}$, where the noise terms ε are independent, with mean 0 and variance σ^2. A forecaster that knows this process has a single model, so (1) must hold for it. Take $h = 2$ and look at the forecast mean. *Direct forecast.* Since $y_{t+2} = \phi y_{t+1} + \varepsilon_{t+2}$ and y_{t+1} is on average ϕy_t, the two-step forecast has mean $\phi^2 y_t$. *Chained forecast.* The one-step forecast is centred on ϕy_t, so the possible next values y are centred there. From the extended context $c \oplus y$ the one-step forecast is centred on ϕy. Averaging ϕy over y gives $\phi \cdot \phi y_t = \phi^2 y_t$. The two means agree. With Gaussian noise the whole distributions agree too: both are normal with mean $\phi^2 y_t$ and variance $(1 + \phi^2)\sigma^2$. A forecaster that predicted a different mean or spread at $h = 2$ than its own one-step forecast implies would fail this check.

[1] Deterministic, Markov, regime-switching and partly observed processes can all be written this way, so the test assumes nothing about which kind produced the data.

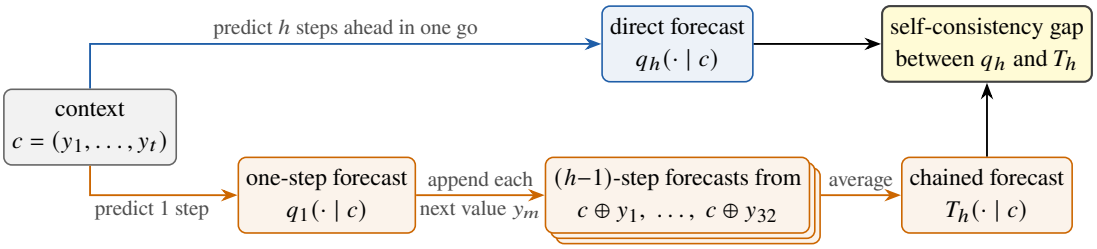


Figure 1: The test. Top (blue): the model's direct forecast h steps ahead. Bottom (orange): the chained forecast, built only from the same model's own forecasts, by predicting one step, appending each of 32 possible next values y_m, forecasting the remaining $h - 1$ steps from each extended context and averaging. A forecaster with a single model of the process has zero gap.

A forecaster is *self-consistent across horizons* if (1) holds. Its *exact self-consistency gap* $\Delta_h(c)$ is the distance between the two sides of (1) as distributions; I use the Wasserstein-1 distance, the average distance between matching quantiles over all quantile levels. The gap uses only the forecaster's own outputs: no future values and no knowledge of the true process. What I measure is a cruder version of Δ_h, defined in Section 2.4.

By *fitting temporal patterns* I mean learning each output, for each horizon and context, to minimise its own expected loss. Nothing in that objective ties the horizons together.

2.2 Why a pattern fitter can fail it

The exact Bayesian forecaster under any prior is self-consistent, because its forecasts are conditionals of one joint distribution, the prior-weighted mixture of the candidate processes, to which the law of total probability applies. Equation (1) appends to the full history, as in my main experiments; a model with a fixed-length window as its state would need its own update rule.

A forecaster that is optimal at every horizon can still fail, as the next example shows.

Example 2 (The patchwork forecaster)**.** Let y be a stationary Gaussian AR(2) process, $y_{t+1} = \phi_1 y_t + \phi_2 y_{t-1} + \varepsilon_{t+1}$, with variance $\gamma(0)$ and autocorrelations $\rho(k)$, the correlation between values k steps apart. The *patchwork* forecaster uses only the last value y_t and, separately for each h, predicts y_{t+h} by $\mathcal{N}\big(\rho(h)\, y_t,\ \gamma(0)(1 - \rho(h)^2)\big)$. For each horizon on its own, this is the best forecast among those that use only the last value. Take $h = 2$ and look at the forecast mean, as in Example 1. *Direct forecast.* The two-step forecast has mean $\rho(2)\, y_t$. *Chained forecast.* The one-step forecast is centred on $\rho(1)\, y_t$. From the extended context $c \oplus y$ the forecaster again uses only the last value, now y, so its one-step forecast is centred on $\rho(1)\, y$. Averaging over y gives $\rho(1)^2 y_t$. The two means differ by $\big(\rho(2) - \rho(1)^2\big) y_t$. The standard AR(2) autocorrelation equations (Yule–Walker), $\rho(1) = \phi_1/(1 - \phi_2)$ and $\rho(2) = \phi_1\rho(1) + \phi_2$, give

$$\rho(2) - \rho(1)^2 = \phi_2 \left(1 - \rho(1)^2\right),$$

which is non-zero whenever $\phi_2 \neq 0$; it is 0.122 for $\phi = (0.5, 0.4)$, where $\rho(1) = 0.833$. The direct two-step forecast is fitted to how y_{t+2} relates to y_t in the data, which reflects both lags of the process. The chained forecast applies the one-lag rule twice, which does not.

So being optimal at each horizon, for the information used, does not make a forecaster self-consistent. Exact agreement with the true conditional distributions at every context and horizon would, because they come from one process, but ordinary per-horizon accuracy scores, however good, do not. This is how the test separates a single model of the process from separate fits for each horizon.

2.3 What a pass and a fail mean

I take understanding the dynamics to include, at least, having a single model of the process, from which self-consistency follows by the law of total probability. So a model that fails lacks this capability in its

outputs, whatever the true process is and whether or not the model is right. A pass shows much less: it rules out separate fits for each horizon and nothing more, since a model that rolls out its own one-step forecast passes however poor that forecast is, because, given the first value y, the rest of its rollout from c is its rollout from $c \oplus y$, which is exactly (1). The test therefore gives evidence about understanding in one direction only. A forecaster that is self-consistent and accurate has a single model that fits the data it was tested on, and no more.

The brief also asks how one would know that a model had succeeded. On synthetic processes, where the exact forecast distribution is known, a successful model should have (i) a gap at the level of a self-consistent baseline, (ii) small error against the exact distribution, and (iii) accurate what-if forecasts, that is, forecasts from $c \oplus y$ for a chosen y. I measured (i) and (ii) for six models on four processes, and (iii) for three models on one process (Section 2.6). No model met (i).

I chose this test over tests on new conditions, such as change points: inside the training distribution a pattern fitter passes those too, and outside it a failure may reflect missing knowledge rather than a missing model.

2.4 How the gap is measured

The models output the quantiles $Q_h(u \mid c)$ of each forecast at the nine levels $u_k = k/10$, $k = 1, \ldots, 9$. I turn each forecast into a full distribution by linear interpolation between these quantiles, with Gaussian tails beyond them. The integral in (1) is replaced by an equal-weight average over 32 evenly spaced quantiles of the one-step forecast, a deterministic version of sampling y from $q_1(\cdot \mid c)$:

$$\widehat{T}_h(\cdot \mid c) = \frac{1}{32} \sum_{m=1}^{32} q_{h-1}(\cdot \mid c \oplus y_m), \qquad y_m = Q_1\Big(\tfrac{m-1/2}{32} \;\Big|\; c\Big).$$

Using 64 values instead of 32 changes the gaps of Chronos-Bolt small and Timestral on AR(2) and electricity by at most 0.0001. With $\widehat{Q}_h(u \mid c)$ the quantiles of $\widehat{T}_h(\cdot \mid c)$, the *measured gap* is

$$D_h(c) = \frac{\frac{1}{9}\sum_{k=1}^{9} \big|Q_h(u_k \mid c) - \widehat{Q}_h(u_k \mid c)\big|}{Q_h(0.9 \mid c) - Q_h(0.1 \mid c)}.$$

The numerator approximates the average distance between matching quantiles over the central 80% of levels. The denominator, the width of the direct 80% interval, makes D_h free of the series' units, so that it can be compared across sources: $D_h = 0.02$ means that the nine quantiles differ on average by 2% of that width. Dividing by a model's own width flatters models with wide intervals, so on synthetic data I also divide by the exact distribution's width, which is the same for every model; on real data there is no common scale, so I do not rank models by D_h there. D_h is not Δ_h, and the interpolation and tails are part of what it measures. A positive D_h shows that the direct and chained central quantiles disagree reproducibly, by more than self-consistent baselines disagree in the same pipeline. It does not show that the full distributions differ outside the nine levels, or that the model has no internal model of the process.

Self-consistent baselines and criterion. The pipeline adds an error of its own, whose size depends on the shape of the forecast. To measure it, forecasters that are self-consistent by construction go through the identical pipeline: on synthetic data the exact forecaster, which outputs the true conditional distribution, and on real data an AR(8) model fitted to each context and then held fixed, once with Gaussian noise and once with noise resampled from its own residuals. The patchwork forecaster of Example 2 is a positive control. The criterion, fixed before any model ran, was that some model exceeds the baseline at some $h \geq 8$ with non-overlapping 95% bootstrap intervals; $h = 2$ and 4 were excluded in advance because the baselines are least reliable there: heavier tails raise the baseline at short horizons (0.0105 at $h = 2$ for t_3 noise), while the Gaussian baselines stay at or below 0.002 at every horizon.

Source	Self-consistent baselines		Models		
	Exact (synthetic) / Gaussian AR(8) (real)	Residual-noise AR(8) (real)	Chronos-Bolt	Timestral-28M	TimesFM 2.5
AR(1)	0.0017	–	0.0167–0.0223	0.0119	0.0075
AR(2)	0.0020	–	0.0185–0.0226	0.0151	0.0083
AR(2), oscillating	0.0015	–	0.0130–0.0157	0.0108	0.0077
AR(1), t_3 noise	0.0012	–	0.0188–0.0229	0.0124	0.0146
Electricity price	0.0015	0.0012	0.0332–0.0421	0.0288	0.0420
COVID-19 cases	0.0016	0.0040	0.1027–0.1150	0.0326	0.0723
Air temperature	0.0017	0.0017	0.0142–0.0186	0.0123	0.0202

Table 1: The measured gap D_h at $h = 16$, mean over 256 contexts. Chronos-Bolt: range over four sizes. Every model's 95% interval lies above every baseline's at every $h \geq 8$.

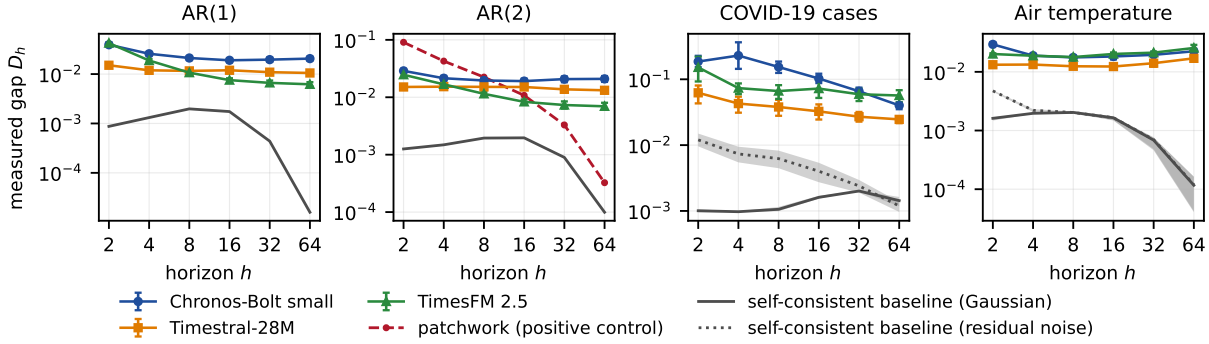


Figure 2: D_h against horizon (log scales), one model per family. Grey: self-consistent baselines with 95% bands (solid: exact forecaster or Gaussian AR(8); dotted: residual-noise AR(8)). Red: the patchwork forecaster. Error bars: 95% bootstrap intervals over contexts (synthetic) or whole series (real).

Data and models. Four synthetic AR sources with known exact distributions and three real datasets from the TIME forecasting benchmark, with 5 electricity markets, 9 countries' COVID-19 cases and 6 weather stations (Table 1); each source has 256 windows of 500 points. The six direct TSFMs come from three families: Chronos-Bolt at four sizes from 9M to 205M parameters [2], Timestral-28M [9] and TimesFM 2.5 (231M). Chronos-T5 [1] was tested separately (supplement).

2.5 Results

Every one of the six models exceeds the self-consistent baselines on every one of the seven sources at every $h \geq 8$ (Table 1, Figure 2), far more than the pre-declared criterion required. On real data each model is above both baselines in every single series, by at least 4.4 times at $h = 8$ to 32, and 6–35 times the residual-noise baseline at $h = 8$ and 16. Dividing instead by the exact distribution's width, which is the same for every model, gives gaps of 0.6–3.8% of that width at $h \geq 8$ on synthetic data, still above the baselines. Size does not help within Chronos-Bolt: there is no monotone trend across its four sizes. Its large gap on COVID is partly a matter of scale: its COVID forecasts are 2.5–5.5 times narrower than elsewhere, because 500-day contexts span whole epidemic waves (supplement).

Two controls show that the test detects what it should and that the gap is not an artefact of the input. The patchwork forecaster is detected, and at $h = 8$ on AR(2) the models' gaps are as large as its own (0.020 for Chronos-Bolt small against 0.022), although at $h = 2$ its gap is three times theirs. Appending a value also lengthens the input and, for patch-based models, shifts the patches; dropping the oldest point whenever a value is appended removes this and changes the gap of Chronos-Bolt small and base and Timestral by at most 6% on four sources (not run for the other three models).

The failure does not depend on fixed dynamics. In a two-regime Markov-switching AR(1), where a

hidden regime sets the AR parameters and 90% of windows contain a switch, the exact Bayesian forecaster (using the Hamilton filter) stays self-consistent within the pipeline's error, while Chronos-Bolt and Timestral stay about 10 and 6 times above it, as without switching (one design, three models; supplement, Table S1).

Generating one step at a time is not enough either. Chronos-T5 also fails on the two Gaussian AR sources where I tested it: at $h = 16$ its excess over the baseline is similar to Timestral's (with every token sampled, a setting chosen after seeing that the default sampler's intervals were too narrow; supplement). A plausible reason is that it reads its own sampled values through its decoder at the original scale, whereas the chained forecast rescales $c \oplus y$ and reads it through the encoder; the runs do not identify which difference matters.

2.6 The gap and accuracy

On synthetic data, where the exact distribution is known, the gap can be compared with each forecast's error against it, all measured as in D_h but divided by the exact distribution's width. The gap can be no larger than the sum of the direct and chained forecasts' errors, and at $h \geq 8$ it is only 4–12% of that sum: most of the models' error is shared by both forecasts. Across models the two agree: ranking the six models by mean gap closely matches ranking them by error (Spearman correlation +0.66 to +0.94 at $h = 8$ and 16), so the more self-consistent models are also the more accurate ones. Within a model the link across contexts is weak (median Spearman +0.18). On real data, where only the realised error (CRPS) is available, neither the ranking nor a per-window link was confirmed (per-window rank correlations at most 0.18 in size, with mixed signs). The gap also shows in what-if forecasts, the forecasts from $c \oplus y$ for a chosen y that the chained forecast is built from: on AR(1), three models' forecasts for the step after an unusual next value (outside the central 90% of the one-step forecast) are about a fifth further from the exact conditional distribution than after a typical one. These analyses were added after the main results, without a pre-declared criterion (supplement).

2.7 How would one build a self-consistent model?

Self-consistency can be trained for, using only the model's own predictions as targets. Let $\rho_u(r) = r\left(u - \mathbf{1}\{r < 0\}\right)$ be the pinball (quantile) loss, whose expected value is smallest at the u-quantile, and let $Q_h^\theta(u_k \mid c)$ be the direct quantiles of a model with parameters θ. Quantile TSFMs such as Chronos-Bolt are trained with a loss like the first term below, on observed futures y_{t+h}; I propose adding the second:

$$L(\theta) = \sum_h \sum_{k=1}^{9} \rho_{u_k}\left(y_{t+h} - Q_h^\theta(u_k \mid c)\right) + \lambda \sum_{h \geq 2} \sum_{k=1}^{9} \mathbb{E}_{z \sim \bar{T}_h(\cdot \mid c)}\, \rho_{u_k}\left(z - Q_h^\theta(u_k \mid c)\right). \tag{2}$$

Here $\bar{T}_h$ is the model's own chained forecast, computed with the current parameters and held fixed, so that no gradient flows through it. The second term is smallest when the direct quantiles equal those of the chained forecast, that is, when (1) holds at the nine levels. It needs no extra data, and $\lambda \geq 0$ sets the trade-off between fitting the observed futures and self-consistency. Holding $\bar{T}_h$ fixed makes the term move the direct forecasts towards the chained ones rather than the reverse.

3 Related work

Equation (1) is classical; for Markov chains it is called the Chapman–Kolmogorov equation. Falck et al. [6] use it to test in-context learning of language models on exchangeable data, averaging the predictive distribution over the model's own imputed samples. Klötergens et al. [7] test the law of total probability in tabular foundation models, but their two sides condition on different data, so even an exact Bayes predictor can fail (as the non-peer-reviewed preprint of Chen and Yeh [4] shows); in my test both sides share the context. Shikhman [11] checks whether deterministic learned simulators give the same answer

when one long step is replaced by several short ones. Econometrics compares direct and iterated multi-step forecasts for accuracy [5, 8]. Baron et al. [3] note that multi-step TSFMs output only per-horizon marginal distributions, and Wang et al. [13] train forecasters so that predictions from true and from self-generated inputs agree. In this lab, Stankeviciute et al. [12] give multi-horizon prediction intervals with joint coverage, and Piskorz et al. [10] elicit one-step predictive distributions from language models; neither tests self-consistency across horizons. The identity and these facts about it are not new. I found no prior audit that applies this test to pretrained probabilistic TSFMs, on synthetic and real series, against self-consistent baselines.

4 Future work and limitations

- **Training for self-consistency.** An open question is whether training with the loss (2) makes a model more self-consistent without costing accuracy, and whether it helps where pattern fitting should fail: long horizons, new regimes and what-if forecasts.
- **Self-consistent by design.** Chronos-T5 failed even though it generates step by step, so generating step by step is not enough. A model that reads its own predictions exactly as it reads data would define one distribution over future paths by construction; the question is whether it can match the direct models' accuracy.
- **A label-free diagnostic.** On synthetic data the gap ranked models as their true error did; on real data this was not confirmed, and it did not flag individual bad forecasts. Knowing when it is informative would make it usable for choosing or monitoring models where no ground truth is available yet, for example after a distribution shift.

Smaller checks remain: separating Chronos-T5's decoder route from its rescaling, the input-length control for all six models, and other interpolations and tails.

Limitations. These models promise accuracy at each horizon, not one distribution over future paths. The test asks whether their outputs nevertheless admit one, which matters whenever forecasts are chained or conditioned on, as in what-if forecasts and sample paths; it does not look inside the models. The evidence with exact baselines comes from simple autoregressive processes, and the real-data evidence from 5 to 9 series per source.

5 Use of AI tools

I used AI tools (Claude in Cursor) for the literature review and the checking of citations; the code, including running the experiments; the figures; and a final language revision, in which the AI rephrased text I had written. AI models also critiqued earlier drafts.

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